Cohort build log (v2 rebuild)

Running record of how each cohort was built, the decisions taken and the numbers produced. Source for the methods section, the flow diagram and the cohort table.

TCGA-LUAD (scripts/10_cohort_tcga.R) — built 19 Oct 2026

- Source: GDC STAR-Counts (GENCODE v36) via TCGAbiolinks 2.40.0; tpm_unstranded, log2(TPM + 1); primary tumours, first barcode per patient. Library type: poly(A) selection.
- Result: 504 patients, 182 deaths (matches the corrected count in the v0.3 HANDOVER).
- A transient "Failure when receiving data from the peer" warning appeared during the GDC metadata query; the cohort was built normally.

CPTAC-3 lung adenocarcinoma (scripts/11_cohort_cptac3.R, R/gdc_api.R) — built 19 Oct 2026

- Why not TCGAbiolinks: GDCquery() stops on CPTAC-3 with "object 'sample_type' not found" (the GDC split sample_type into tissue_type, tumor_descriptor, specimen_type and preservation_method; some CPTAC-3 samples carry only the new fields). Its barcode filter also compares against aliquot IDs, so it could not have matched CPTAC-3 case IDs. CPTAC-3 is therefore read directly from the GDC API.
- GDC Data Release 46.0 (10 Aug 2026).
- Cases: 1,866 CPTAC-3; 339 lung; 230 lung adenocarcinoma (primary_site "Bronchus and lung" and disease_type "Adenomas and Adenocarcinomas"); all "Adenocarcinoma, NOS", one diagnosis each.
- RNA-seq files of these cases: 446 (215 normal, 3 metastatic, 228 primary tumour). 50 primary-tumour files list several samples of the same patient (one RNA extract from several tumour pieces; the CPTAC LUAD study cryopulverised specimens and aliquoted the material); none was pooled across patients. Rule: a file is eligible if it comes from one case and every sample is a primary tumour; one file per patient (first by sample ID, then file ID).
- Patients with RNA-seq: 228; with valid survival: 205; deaths: 55. Exclusions for survival are mainly vital status "Not Reported" (22 of 230 cases).
- Last follow-up: latest follow_ups $days_to_follow_up (the rule TCGAbiolinks applies to TCGA); $diagnoses days_to_last_follow_up only as a fallback. Checked: the two sources are identical for 204 of 205 analysed patients; the one difference (63 days) is a patient who died and has a follow-up record dated after death, which does not affect overall survival (death date used).
- Age: age_at_index is filled for only 2 of 230 cases; age is taken from diagnoses$age_at_diagnosis (days, converted to completed years) for 225, so age is recorded for all 205 analysed patients. This keeps age in the H2 model (used when at

least 80% recorded), as for TCGA.

- Stage: I 114, II 46, III-IV 45 (all staged). Sex: 131 male, 74 female. Median follow-up (reverse Kaplan-Meier) 35.6 months; maximum 72.
- Change from v0.3: the old CPTAC-3 cohort had 150 patients and 38 deaths; the old filter most likely dropped the files that list several tumour samples. All CPTAC-3 numbers in the manuscript will change.

RNA-seq library type and TPM composition (TCGA-LUAD vs CPTAC-3) — 19 Oct 2026

- CPTAC LUAD libraries: TruSeq Stranded Total RNA with Ribo-Zero Gold (total RNA, rRNA-depleted; Gillette et al., Cell 2020, doi:10.1016/j.cell.2020.06.013). TCGA: poly(A) selection. Many lncRNAs are not polyadenylated or stay in the nucleus, so the two library types capture them differently: a cohort-level source of heterogeneity for the Discussion and the cohort table.
- Share of TPM outside protein-coding and lncRNA genes, per sample (min / Q1 / median / Q3 / max): TCGA-LUAD 0.017 / 0.062 / 0.095 / 0.173 / 0.748; CPTAC-3 0.273 / 0.367 / 0.413 / 0.468 / 0.672.
- Median log2(TPM + 1), TCGA vs CPTAC-3: RN7SL1 9.19 vs 17.81; MT-RNR2 13.17 vs 12.63; GAPDH 10.49 vs 9.73; ACTB 12.01 vs 10.74.
- Within-cohort z-scoring removes the constant shift but not the sample-to-sample part (interquartile spread about 0.18 log2 units in TCGA and 0.25 in CPTAC-3; the most extreme TCGA samples are about 2 log2 units low).
- Decision: primary analysis unchanged (GDC TPM as provided). Pre-specified sensitivity analysis added to the kit (config sensitivity$tpm_rescale_gene_types; rescale_tpm_cohort(); setting in scripts/34_sensitivity.R; QC columns in scripts/19_cohort_qc.R): RNA-seq TPM rescaled so that protein-coding and lncRNA genes (GENCODE v36) sum to 10^6 per sample before log2(TPM + 1) and z-scoring. Must be written into the OSF amendment before any outcome analysis. No sample exclusion based on this share (it would be data-driven); the share is reported in QC.

GEO microarray cohorts (scripts/12_cohort_gpl570.R, config/cohorts.yaml) — settings fixed 19 Oct 2026

Fields confirmed against logs/geo_fields.txt (show_geo_fields output). An event value that is neither dead nor alive (e.g. "NTL", "n/a") is coded missing, never alive.

- GSE31210: tissue "primary lung tumor" (226 of 246; 20 normal lung excluded). OS = "days before death/censor" (days); event "death" dead/alive (35 dead, 191 alive). Stage "pathological stage" IA/IB/II. Age from "age (years)", completed from "age" (GEO splits age over two fields; the second holds 5 samples). Sensitivity flag: "exclude for prognosis analysis due to incomplete resection or adjuvant therapy" = "exclude" (22 cases).
- GSE50081: histology "adenocarcinoma" only (127 of 181; adenosquamous, large cell, "NSCLC-favor adenocarcinoma" excluded). OS = "survival time" (years); event "status" dead/alive. Stage "Stage" is written 1A/1B/2A/2B and is recoded to IA/IB/IIA/IIB (without

the recode every stage would have been missing). Age "age"; sex "Sex".

- GSE30219: histology "ADC" (85 of 307). OS = "follow-up time (months)"; event "status" DEAD/ALIVE. No stage field: stage group derived from pT/pN/pM with AJCC 6th-edition grouping (T not split into a/b in this series): I = T1-2 N0 M0; II = T1-2 N1 M0 or T3 N0 M0; III-IV = T3 N1-2, T4, N2-3 or M1; MX or missing M counted as M0 (resected series); TX/NX give missing stage unless III-IV anyway. Limitation to state: under AJCC 7th edition some T2 N0 tumours (>5 cm) would be stage II.
- CEL archives: downloaded under a temporary name and kept only when the size matches the server's; RMA stops if any selected sample lacks a CEL file. BioMart requests (Ensembl release 114 probe map) are retried.
- First run of script 12 (19 Oct 2026): GSE50081 127 patients, 51 deaths, median follow-up 65.9 months, stage I 92 / II 35; GSE30219 83 of 85 with valid survival, 43 deaths, median follow-up 116 months, but stage missing for all 83. Cause: the R yaml package reads an unquoted key `n` as FALSE (YAML 1.1 booleans y/n/yes/no/on/off), so `stage_tnm: {t, n, m}` lost the N field. Fixed by naming the keys tumour / node / metastasis; make_clin_geo() now stops if any is missing, and the config test reads the real YAML file. Rule for all config files: no bare y/n/yes/no/on/off keys or values.
- GPL570 probe map: 54,675 probes; 20,274 Ensembl genes kept (one probe per gene).
- Accepted (19 Oct 2026): GSE31210 226 patients, 35 deaths, median follow-up 60.7 months, stage I 168 / II 58, age for all 226 (two GEO age fields combined), 22 flagged for the sensitivity analysis; GSE50081 127 patients, 51 deaths, median follow-up 65.9 months, stage I 92 / II 35 (1A 36, 1B 56, 2A 7, 2B 28). Expression range 1.85-15.08 (RMA log2).
- GSE30219: 85 adenocarcinomas, 83 with valid survival, 43 deaths, median follow-up 116 months. Follow-up is recorded in whole months; the 2 excluded patients have follow-up 0 and both died (deaths in the first month after surgery are coded 0 and fall under the pre-specified rule follow-up > 0): state in the flow diagram. Stage after the fix (20 Oct 2026; 85 selected): T1N0M0 71 and T2N0M0 10 -> I; T2N1M0 2 and T3N0M0 1 -> II; T3N1M0 1 -> III-IV. Analysed 83: I 79, II 3, III-IV 1. Stage is nearly constant in GSE30219 (and GSE31210 / GSE50081 hold only stage I-II by design), so stage adjustment in H2 carries little information in the microarray cohorts: state this when reporting H2.
- GSE81089 (RNA-seq, 199 NSCLC + 19 paired normal; Uppsala University Hospital, surgery 2006-2010; clinical data from the regional lung cancer registry). Codebook from the series matrix header: histology 1 = squamous cell, 2 = adenocarcinoma ("AC unspecified"), 3 = large cell/NOS (2 = 108 samples); stage (pTNM, reviewed 2013) 1 = IA, 2 = IB, 3 = IIA, 4 = IIB, 5 = IIIA, 6 = IIIB, 7 = IV; dead 0 = no, 1 = yes; vital date = date of death or latest contact; surgery date = date the sample reached pathology; age at surgery; smoking 1 = current, 2 = ex (>1 year), 3 = never. Overall survival = vital date minus surgery date. Expression: GSE81089_FPKM_cufflinks.tsv.gz and GSE81089_readcounts_featurecounts.tsv.gz (columns L608T_2122 and L771T_1 belong to samples L608T and L771T). The v0.3 pipeline dropped this cohort because of an identifier bug (R9), not missing data; restoring it requires the OSF amendment. Library (Djureinovic et al., JCI Insight 2016, doi:10.1172/jci.insight.86837): Illumina TruSeq RNA

v2 with poly(A) selection (as TCGA), HiSeq 2500, 2 x 100 bp; TopHat 2.0.8b on GRCh37 with Ensembl 73 genes; FPKM from Cufflinks 2.1.1. FPKM file: 63,130 Ensembl genes (no version suffix) x 218 samples named as in the GEO field "tumor (t) or normal (n)". Script 13: adenocarcinomas (histology 2), TPM = FPKM / sample total x 1e6, log2(TPM + 1); follow-up from the two dates. Ensembl 73 is older than GENCODE v36, so newer lncRNA IDs will be absent: coverage will show it. Built for description; enters the analyses only after the amendment. First run stopped: the FPKM file contains missing values. Rule added: genes with a missing FPKM in any selected adenocarcinoma are dropped (they could not pass the presence rule anyway) and TPM is computed over the remaining genes; the script stops if any sample has more than 5% missing. Second run stopped on negative values: all 579 negative FPKM values among the adenocarcinomas are exactly -1 (a code for "not quantified"; at most 13 per sample, at most 52 samples per gene), so -1 is treated as missing; any other negative value stops the script. Dropped genes are listed in data/processed/GSE81089_genes_dropped_missing.csv, to check against the signature genes at the coverage step. Counts reported in results/qc/GSE81089_build_summary.txt.

- GSE81089 built (20 Oct 2026): histology codes of all samples 1 = 67, 2 = 108, 3 = 24, missing 19 (normals). 108 adenocarcinomas: 49 dead, 59 alive; stage codes 1:44, 2:18, 3:8, 4:11, 5:24, 7:3. FPKM file 63,130 genes x 218 samples; among the adenocarcinomas 579 values coded -1 (75 genes) and 1 blank (in a non-Ensembl row "TC%..."); 76 rows dropped, 63,054 Ensembl genes kept (all kept rows are Ensembl genes, so the stray row never entered the TPM totals). Analysed: 107 patients (1 excluded: follow-up 0, died on the day of surgery), 48 deaths, median follow-up 52.3 months (registry extraction dates 28 Apr 2013 and 10 Apr 2013 dominate the last-contact dates), stage I 61 / II 19 / III-IV 27, age for all, 69 female / 39 male. Expression range 0-17.62. Listed in config as cohorts$candidate (QC and coverage only) until the amendment adds it.

All cohorts built (20 Oct 2026)

Cohort	Role	Platform / library	Patients	Deaths	Median follow-up (months)
TCGA-LUAD	reference (apparent performance)	RNA-seq, poly(A)	504	182	29.0
CPTAC-3	independent	RNA-seq, total RNA (Ribo-Zero)	205	55	35.6
GSE31210	independent	GPL570, RMA	226	35	60.7
GSE50081	independent	GPL570, RMA	127	51	65.9
GSE30219	independent	GPL570, RMA	83	43	116
GSE81089	independent (restored; must be named in the amendment)	RNA-seq, poly(A), Ensembl 73	107	48	52.3

Independent deaths:
184 without
GSE81089, 232 with it.

- Open gap: config lists GSE72094 as coverage_only, but no script builds it. Its coverage can be judged from the GPL15048 platform annotation; to be added at the coverage step (Step 3).

Cohort QC (scripts/19_cohort_qc.R) — 20 Oct 2026

- All six cohorts pass check_cohort(). Sample flow (with expression / with clinical record / with valid survival): TCGA-LUAD 517 / 517 / 504; CPTAC-3 228 / 228 / 205; GSE31210 226 / 226 / 226; GSE50081 127 / 127 / 127; GSE30219 85 / 85 / 83; GSE81089 108 / 108 / 107.
- Stage I-II: TCGA 78.6% (8 missing), CPTAC-3 78.0%, GSE31210 100%, GSE50081 100%, GSE30219 98.8%, GSE81089 74.8%. Age recorded: TCGA 98%, all others 100%.
- Share of TPM outside protein-coding and lncRNA genes (median / max): TCGA 0.095 / 0.748; CPTAC-3 0.413 / 0.672; GSE81089 0.251 / 0.555 (Ensembl 73 IDs matched to GENCODE v36 gene types, so retired IDs count as "other").
- Expression boxplots: the three GPL570 cohorts are aligned after RMA; RNA-seq panels show all ~60,000 genes (medians near 0) and nothing abnormal; one GSE81089 sample has a lower upper quartile, consistent with the composition differences.
- Decision (20 Oct 2026): GSE81089 included as an independent cohort (config cohorts$independent; script 13 in run_all.R). The OSF amendment must name it before scripts 30-34 run, and independence.csv must be completed for it.

- Project folder moved by the user to ~/Desktop/Lab Works/MHENCR/luad-lncrna-benchmark-v2 (20 Oct 2026). All paths in the pipeline are relative to the project root, so nothing in the code changes.

After OSF Update 3 (filed 20 Oct 2026)

- OSF Update 3 filed and committed by the user on 20 Oct 2026 (field-by-field text: claude/osf-update3-final-paste-text.md).
- Project folder moved by the user to ~/Desktop/Lab Works/BenchMark3/luad-lncrna-benchmark-v2 (20 Oct 2026). Paths in the pipeline are relative, so no code changes.
- Code update v2-update-D2-D4 (20 Oct 2026):
  - D2: stage coded in four levels I / II / III / IV everywhere (collapse_stage(); stage_from_tnm() puts M1 in IV, other III-defining TNM in III).
  - D4: TCGA-LUAD keeps, per patient, the primary tumour with the largest total read count (sum of STAR unstranded counts over Ensembl genes); ties by barcode. choose_tcga_samples(); choices written to data/processed/TCGA-LUAD_sample_choice.csv. Script 10 reuses the saved SummarizedExperiment and clinical data (no new download) and keeps the GDC release of the first download.
  - CPTAC-3 frozen: script 11 reuses the saved GDC answers (cases_api.rds, files_api.rds) so the cohort stays at Data Release 46.0; querying again would be a reportable deviation.
  - Script 21 downloads GENCODE v22 from the EBI GENCODE archive if missing; *.gtf.gz kept out of git.
  - independence template extended with GSE72094 rows (15 x 7 cohorts).
  - All cohorts must be rebuilt once (scripts 10, 11, 12, 13, 19) because stage is stored in the cohort objects. Patients and deaths are expected unchanged; only stage III/IV split and the TCGA sample choice change.

Rebuild with D2/D4 checked (20 Oct 2026)

- All cohorts rebuilt by the user; patients and deaths unchanged (TCGA 504/182, CPTAC-3 205/55, GSE31210 226/35, GSE50081 127/51, GSE30219 83/43, GSE81089 107/48). TCGA-LUAD stage I 270 / II 120 / III 81 / IV 25 / missing 8. TCGA: 601 RNA-seq files, 540 primary-tumour samples from 517 patients; 12 patients had several primary-tumour samples and the read-count rule changed the chosen sample for 1 (TCGA-44-5645: -01A-01R-A278-07 instead of -01A-01R-1628-07). GDC Data Release 46.0 kept from the first download. GENCODE v22 downloaded: 60,483 genes (v36: 60,616). Commit 3b3ab1b (an unzipped update folder was committed by mistake and removed in the next commit).

Analysis-code package (v2-update-analysis, 20 Oct 2026)

Implements the filed plan (Update 3, D1-D6) and the registered analyses that were missing. Implementation choices fixed before any outcome analysis (state them in the Methods):

- Positive control (script 29, all cohorts): Cox model on stage factor (I reference).

Registered rule when >= 10 stage III-IV patients: HR(III vs I) > 1, Wald p < 0.05, HR non-decreasing across the stage levels present (1 <= HR II <= HR III <= HR IV). D1 adaptation below 10: HR(II vs I) > 1, p < 0.05 (stage III/IV levels ignored for monotonicity). Failing cohorts are flagged "exploratory" in scripts 30-32.

- H1 (script 30): random-gene pool = measurable genes of GENCODE v36 type protein_coding or lncRNA (D3); signature genes kept only to place them in their decile. Cox-model C-index reported beside the primary C-index.
- H2 (script 31): Model A = stage (I-IV) + sex (male reference) + age (if recorded >= 80%); complete cases for the covariates in the model; LRT 1 df; Holm within cohort (D5); delta C-index with 1,000-resample percentile bootstrap (both models refitted). The non-registered "at least 30 deaths" condition was removed (it never applied).
- Per-pair metrics (script 32): time-dependent AUC = cumulative/dynamic IPCW estimator (Uno 2007; censoring Kaplan-Meier, cases weighted 1/G(T-)), at 12 and 36 months, 1,000-resample percentile CI; log-rank test = score above the within-cohort median vs at or below; cox.zph on Surv ~ scale(score), p < 0.05 -> RMST difference (high minus low median group) with tau = the smaller of the two groups' maximum follow-up (survRM2 default), SE from survfit; pooling only with >= 3 independent cohorts (D6), REML + Hartung-Knapp, leave-one-cohort-out; 2 cohorts shown unpooled.
- Direction consistency (script 35): clusters = connected components of signatures with Jaccard > 0.3 (Ensembl gene sets from gene_map.csv); one observation per cluster per independent cohort = mean log-HR per SD of the cluster's evaluable signatures; consistent = mean > 0; two-sided binomial test vs 50%, reported at the effective (cluster x cohort) and nominal (pair) level.
- Gene-level concordance (script 35): univariable Cox on each measurable signature gene's z-scored expression, sign vs published coefficient sign; all cohorts, flagged by independence and evaluability; summaries over independent cohorts.
- Power (script 36): normal score, exponential event times with hazard exp(beta x), beta set analytically for true C = 0.60/0.65/0.70, exponential censoring set to each cohort's share of deaths; test of C = 0.5 from survival::concordance variance; alpha 0.05 and 0.05 / evaluable signatures (strictest Holm step); 1,000 replicates. Upper bound for H1 (the random-gene null is wider than chance).
- Sensitivity (script 34): measurable genes recomputed in every setting; H1 rerun for lncRNA-only and all-gene null pools, stage I-II only, early deaths excluded, GSE31210 without the 22 flagged cases; transportability rerun for every setting (B = 1000).
- GSE72094 (script 14): first run writes logs/gse72094_fields.txt and stops; builder maps GPL15048 probes by Ensembl or Entrez ID (Entrez via Ensembl 114 BioMart, one-to-one only), never by symbol; one probe per gene (highest mean). Moves into cohorts$independent once built.

Stage positive control (script 29) - 20 Oct 2026

All six cohorts pass. TCGA-LUAD (I 270 / II 120 / III 81 / IV 25): registered rule, HR(III vs I) 3.49, p = 1.6e-10, monotonic, stage-only C 0.664. CPTAC-3 (I 114 / II 46 / III 43 / IV 2): registered,

HR(III vs I) 2.47, p = 0.008, monotonic (HR II vs I 1.88, p = 0.058), C 0.625. GSE81089 (I 61 / II 19 / III 24 / IV 3): registered, HR(III vs I) 4.22, p = 4e-5, monotonic, C 0.677. Adapted rule (D1): GSE31210 (I 168 / II 58) HR(II vs I) 4.23, p = 2e-5, C 0.678 (same as the stage-only C-index disclosed from v0.3); GSE50081 (I 92 / II 35) 2.44, p = 0.002, C 0.591; GSE30219 (I 79 / II 3 / III 1) 4.45, p = 0.015, C 0.541 (the single stage III patient was censored: HR(III vs I) ~0, not used). Tests passed on the user's Mac before the run.

GSE72094 fields (logs/gse72094_fields.txt, 20 Oct 2026)

442 samples (one per patient, K001...), 60,607 probes, expression already log2 (-0.31 to 15), no missing values. survival_time_in_days (44 "NA"), vital_status Alive 298 / Dead 122 / NA 22, Stage 1A 158, 1B 101, 2A 18, 2B 51, 3A 43, 3B 19, 4 17, 1 6, 3 1, NA 28; age_at_diagnosis; gender F 240 / M 202. GPL15048 annotation: ID, GB_LIST, EntrezGeneID (41,090 of 60,607 probes), GeneSymbol, SPOT_ID. Probes mapped by Entrez ID to Ensembl (release 114, one-to-one only), never by symbol; one probe per gene (highest mean). GSE72094 moved into cohorts$independent (config) with this build.

GSE72094 built (20 Oct 2026) and probe maps restricted to the primary assembly

- First build: 442 samples, 398 with valid survival (44 without survival time), 113 deaths, median follow-up 29.9 months (max 68.2), stage I 254 / II 67 / III 57 / IV 15 / missing 5, age 421 of 442, F 240 / M 202. Probes: 41,090 with one Entrez ID; Entrez IDs 18,158 one-to-one, 1,390 to several Ensembl genes (dropped), 2,595 to none; 18,136 genes kept. Expression 0.37-14.51 (log2, as submitted).
- Problem found: Ensembl BioMart also lists gene copies on alternative haplotypes/ patches (e.g. MHC), so a probe or Entrez ID hitting such a copy counted as "several genes" and was dropped. Signature genes come from GENCODE primary-assembly tables, so these copies can never match a signature gene; dropping the probe could make a present gene look absent. Fix (before any outcome analysis): both maps (GPL570 probe -> Ensembl; Entrez -> Ensembl) keep only genes on chromosomes 1-22, X, Y, MT (keep_primary()); new cache files gpl570_probe_map_ensembl_primary.csv and entrez_to_ensembl_e114_primary.csv. GPL570 cohorts and GSE72094 to be rebuilt (scripts 12, 14, 19); patients and deaths unchanged, gene counts expected to rise. State in Methods: probes mapped to primary-assembly Ensembl genes only.
- Rebuilt with primary-assembly maps (20 Oct 2026): GPL570 probe-gene pairs 49,996; genes kept 21,407 (was 20,274) in GSE31210, GSE50081 and GSE30219; patients/deaths unchanged (226/35, 127/51, 83/43). GSE72094: Entrez IDs one-to-one 19,468, several 70 (was 1,390), none 2,605; 37,665 probes -> 19,445 genes (was 18,136); 398 patients, 113 deaths unchanged.
- All seven cohorts now built. Independent deaths: CPTAC-3 55, GSE31210 35, GSE50081 51, GSE30219 43, GSE81089 48, GSE72094 113 (total 345).
- Waiting on the user: data/signatures/independence.csv (15 x 7) and data/signatures/ signature_master.csv (blind second extraction, annotation v22/v36, DOI, source location, agreed coefficient). Then scripts 01, 22, 23 (gene map, coverage) and review before

scripts 29-36.

Signature extraction and independence (20 Oct 2026)

- Double extraction: extractor 1 = Moinul Islam (values carried from v0.3), extractor 2 = a lab mate working blind from blind_sheet_extractor2 (no access to extractor 1 values). 102 genes in 15 signatures: 86 identical, 16 differing only by rounding (Jin 2020 x7, Li 2023 x7, Yang 2022 x2: extractor 1 rounded); no sign or value disagreement. Final coefficient = full-precision value printed in the paper. Script 01 now classifies agreement (results/tables/extraction_agreement.csv; numbers extraction_identical / _rounding_only / _disagreements) and requires one DOI per signature.
- Corrections on review: DOIs of Yang 2023 (10.1155/2023/6670514) and Zhang 2025 (10.2147/OTT.S538953) had been Excel-autofilled (+1 per row); all 15 DOIs verified against PubMed (first author, year, journal match). Jin 2020 source_location: Table 2 holds the LASSO Cox coefficients used in the risk score (Section 3.2, full text checked), not univariate coefficients.
- `annotation` = the gene table whose naming the paper's gene names follow (not the expression data model; the earlier instruction "v22 if HTSeq/FPKM" was imprecise). Revised: Diao 2022, He 2021, Jin 2020, Song 2021 v22 -> v36 (accession-style names such as AL034397.3, ITGB1-DT); Yu 2024 v36 -> v22 (RP11-21L23.2, RP11-295G20.2, KIAA0125). Stated releases with nearest table: Geng 2021 GENCODE 37 -> v36; Zheng 2021 GRCh38.p12 -> v36; Lin 2024 GRCh37 -> v22; Zhang 2021 GENCODE v7 -> v22. Script 22 now writes gene_map_annotation_check.csv (names each table knows, per signature) and gene_map_conflicts.csv (names with different IDs across tables) for review.
- Independence (15 x 7): non-independent outside TCGA-LUAD: Geng 2021 x GSE31210 (validation); Yu 2024 x GSE31210 (feature selection; validation), x GSE50081 (validation, combined with GSE37745), x GSE30219 (validation); Yang 2022 x GSE50081 (validation of signature gene LINC01214, Fig. 9). All other pairs independent (searched by accession and author/institution names).
- Script 22 first run (20 Oct 2026): the name check confirmed the revised annotations (Diao v22 1/4 vs v36 4/4; He 3/7 vs 7/7; Jin 2/7 vs 5/7; Song 2/5 vs 3/5; Yu 9/9 vs 6/9) and flagged three more, revised to v36: Yang 2022 (3/5 vs 5/5), Zhang 2021 (0/4 vs 4/4, although the paper states GENCODE v7), Lin 2024 (2/5 vs 5/5). Yao 2021 and Wu 2022: all names in both tables. 11 names point to different Ensembl IDs in v22 and v36 (resolved by the paper's annotation; to be reviewed in gene_map_conflicts.csv). Not in either table: AL034397.3 (Jin 2020, Song 2021) and HSPC324 (Jin 2020); scripts/22b_name_history.R looks them up in Ensembl archives 90-104 for the override evidence.
- Conflicts reviewed (20 Oct 2026), all resolved to the v36 ID chosen by the paper's annotation; each signature involved has all its names in v36 and fewer in v22, and every chosen ID is an lncRNA in v36: Yang 2023 AC091132.2 -> ENSG00000236234 (v22 name = ENSG00000264038), U91328.1 -> ENSG00000272462 (v22: ENSG00000281052; also Zhang 2025); He 2021 / Zheng 2021 AC091185.1 -> ENSG00000253476 (v22: ENSG00000277665); He 2021 / Geng 2021 AL590226.1 -> ENSG00000228401 (v22:

ENSG00000221693); Jin 2020 AC091057.1 -> ENSG00000187951 (v22 name belongs to protein-coding ENSG00000280998); Geng 2021 PLAC4 -> ENSG00000280109 (one of the two v22 IDs); Li 2023 AC007384.1 -> ENSG00000237513 (v22: ENSG00000280574); Zhang 2025 AC090559.1 -> ENSG00000255197 (v22: ENSG00000265467). Lesson: clone-based names are reused for different genes across releases, so the naming era must be settled per paper.

- Ensembl BioMart archives 90-104 were not reachable from the user's Mac; script 22b rewritten to use NCBI Gene (org.Hs.eg.db, offline) and intermediate GENCODE GTFs (v33, 32, 31, 29, 27, 35, 38) instead.
- Unplaced names resolved (20 Oct 2026; results/tables/gene_name_history.csv): AL034397.3 (Jin 2020, Song 2021) = ENSG00000274536 (GENCODE v32; chrX:66015461-66020422 +; RP6-159A1.4 in v22, MIR223HG in v36) - the same gene as MIR223HG in Geng 2021 and Lin 2024. HSPC324 (Jin 2020) = ENSG00000228401 (GENCODE v29; NCBI alias of LNCEGFL7OS, Entrez 101928612; RP11-251M1.1 in v22, AL590226.1 in v36) - the same gene as AL590226.1 in He 2021 and Geng 2021. All papers give these genes negative coefficients. Written to config/gene_overrides.csv with evidence. Finding for the paper: matching by Ensembl ID reveals gene sharing between signatures that symbol matching hides (MIR223HG in 4 signatures, LNCEGFL7OS/AL590226.1 in 3). GENCODE v33 download was interrupted (no effect: v32 and v29 found the names).

Coverage (script 23, 20 Oct 2026) - first look

- Evaluable (>= 80%): TCGA-LUAD 14/15 and CPTAC-3 14/15 (Diao 2022 0.75: LINC00941 missing in both); GPL570 cohorts (identical coverage): Yao 2021 6/7 (C20orf197 missing) and Yu 2024 8/9 (CROCCP2 missing; Yu 2024 non-independent there); GSE81089: He, Zheng, Yu, Zhang 2025, Wu, Yang 2022, Yao, Lin (8; newer IDs absent from Ensembl 73); GSE72094: none (max 2/9). AC026355.1 (Zheng, Zhang 2025) measurable in CPTAC-3 but not TCGA-LUAD (expression).
- Independent evaluable pairs: CPTAC-3 14 + Yao 2021 in GSE31210/GSE50081/GSE30219 + GSE81089 8 = 25. Only Yao 2021 reaches >= 3 independent cohorts (5): the only pooled signature.
- To check before analysis (script 23b_presence_check.R): LINC00941 absent in TCGA/CPTAC but present in GSE81089 (suspicious for a known LUAD-expressed lncRNA); C20orf197 absent on GPL570 although v0.3 found probes (possible ID retired between v22 and Ensembl 114).
- Presence check (script 23b, 20 Oct 2026): (1) LINC00941 ENSG00000235884 has zero counts in every TCGA-LUAD and CPTAC-3 sample (GDC STAR, GENCODE v36) although present in GSE81089 (Ensembl 73); Ensembl ID history: last in release 111, replaced by ENSG00000285517 (overlapping model that takes the reads). (2) Nine signature IDs are not in Ensembl 114, so no GPL570/GPL15048 probe can map to them: seven with an official successor (LINC00941 -> ENSG00000285517; LINC00707 ENSG00000238266 -> ENSG00000223784; AC026355.2 ENSG00000228561 -> ENSG00000228221; AP000695.2 ENSG00000233818 -> ENSG00000230479 =

AP000695.1; AL161431.1 ENSG00000275216 -> ENSG00000223617; FRMD6-AS1 ENSG00000273888 -> ENSG00000258537; C20orf197 ENSG00000176659 -> ENSG00000228340 MIR646HG), two retired without successor (AC027031.2 ENSG00000254615, last release 102; AC007384.1 ENSG00000237513, last 104). (3) AC026355.1 (Zheng 2021, Zhang 2025) maps in v36 to ENSG00000200288, an ID range typical of small RNAs; barely expressed in TCGA (3.4%) and GSE81089 (0%), measurable in CPTAC-3 (total RNA): suspected wrong gene.

- Decision (before any outcome analysis): config/gene_id_alternatives.csv lists the seven official successors with evidence; a successor is used in a cohort only where the original ID is missing from that cohort's gene model or not measurable, and only if the successor is measurable (cohort_gene_map(); coverage_table() reports via_alternative_id). Script 22c_name_audit.R compares every name across GENCODE v22, v29, v31, v32 and v36 with gene types, flagging non-lncRNA types and names whose ID changes across releases. State in Methods: "IDs absent from a cohort's gene model were matched to their Ensembl successor (Ensembl ID history)".
- Coverage with successor IDs (20 Oct 2026): Diao 2022 4/4 in TCGA-LUAD and CPTAC-3 (LINC00941 via ENSG00000285517); Yao 2021 7/7 in GSE31210, GSE50081, GSE30219 (C20orf197 via ENSG00000228340 MIR646HG); Wu 2022 6/6 in GSE81089 (FRMD6-AS1 via ENSG00000258537); GSE72094 still none >= 0.8.
- Name audit across GENCODE v22, v29, v31, v32, v36 (results/tables/gene_name_audit.csv). Wrong-gene mappings corrected by override: AC026355.1 (Zheng 2021, Zhang 2025) was ENSG00000200288, a snoRNA in v32-v36; in v29-v31 the name belonged to lncRNA ENSG00000228561 (= AC026355.2 of He 2021 from v32; all three negative coefficients) -> ENSG00000228561. AC010980.2 (Song 2021) was the v22 transcribed unprocessed pseudogene ENSG00000237732; v29 lincRNA ENSG00000267034 -> ENSG00000267034. Accepted as is: miRNA genes that reused names in v22 (AC007384.1, AC090559.1, AC091057.1, AC091132.2, AC091185.1, AL590226.1, U91328.1) - lncRNA IDs already chosen; type relabels only (CROCCP2 lincRNA -> transcribed_unprocessed_pseudogene; AC018629.1 TEC in all releases). Residual uncertainty: AP001453.2 (Diao 2022) is ENSG00000256940 in v29/v31 but ENSG00000257086 in v32/v36; kept v32+ (a 2022 paper), and He 2021 AC026355.2 kept as ENSG00000228561 (its AL590226.1 requires >= v31; v32+ consistent with the publication date and coefficient signs). Hidden overlaps found by ID: MIR223HG/AL034397.3 (4 signatures), LNCEGFL7OS/AL590226.1/HSPC324 (3), AC026355.1/AC026355.2 (3).
- Next: rerun 22, 23, 23b; freeze inputs (git tag v2-preanalysis); then scripts 29-36.
- Coverage after the AC026355.1/AC010980.2 overrides (20 Oct 2026): TCGA-LUAD and CPTAC-3 all 1.0 except Song 2021 0.8 (AC010980.2 = ENSG00000267034, last in Ensembl release 100, no replacement listed; absent from GENCODE v36; present in GSE81089). GSE81089: Zheng 2021 0.9, Zhang 2025 1.0. Independent evaluable pairs (>= 0.8, not used by authors): 26 = CPTAC-3 15 + Yao 2021 in GSE31210, GSE50081, GSE30219 + GSE81089 8 (He 2021, Zheng 2021, Yu 2024, Zhang 2025, Wu 2022, Yang 2022, Yao 2021, Lin 2024). Only Yao 2021 has >= 3 independent cohorts (5): the only pooled signature.

GSE72094: none evaluable. AC026355.1/.2 (ENSG00000228561) has no probe on GPL570 or GPL15048, nor does its successor.

- Independence confirmed (20 Oct 2026, extractors): no paper reports Kaplan-Meier Plotter or a similar portal for survival validation. Paper 62 (Yao 2021) used GEPIA (TCGA/GTEx) only for lncRNA-mRNA expression correlation. Direct GEO validation: Paper 53 Yang 2022 GSE50081 (single gene LINC01214; classed as used, conservative; pair not evaluable anyway, 0.6), Paper 192 Geng 2021 GSE31210, Paper 305 Yu 2024 GSE30219, GSE31210, GSE50081 (with GSE37745). Matches independence.csv.
- Predecessor-ID check added (scripts/23c_predecessor_check.R; R/coverage.R history_walk, predecessor_hits; decision rule fixed 20 Oct 2026, before any outcome analysis): GSE81089 uses Ensembl 73, so 13 signature-gene IDs created later are absent there. Official Ensembl ID history of every retired GSE81089 ID is followed; an older ID is accepted when measurable, its history passes through the signature gene's own ID and ends in exactly one current gene, and it is the only such ID. Accepted IDs go into config/gene_id_alternatives.csv. Same rule as the successor IDs, other direction. Deviation to log in docs/deviations (not in the registration). Candidate signatures one gene short of 0.8 in GSE81089: Geng 2021, Jin 2020, Li 2023, Yang 2023, Song 2021.
- Predecessor check result (24 Oct 2026, scripts/23c_predecessor_check.R): 63,054 GSE81089 IDs queried; 10,024 retired in the current Ensembl release; no retired ID's official history leads to any missing signature gene -> no predecessor IDs accepted; config/gene_id_alternatives.csv unchanged; GSE81089 coverage stands (8 evaluable). Replacement chains were followed (at least one extra lookup round ran), so the null result is not a parsing artefact. For Methods: GSE81089 genes created after Ensembl 73 are counted as not measurable; official ID history checked in both directions.
- Inputs frozen: commit 871d199, git tag v2-preanalysis (24 Oct 2026). Next: scripts 29, 30, 31, 32, 35, 36, 33, 34.
- No git remote (local repository only); the v2-preanalysis tag is local. (24 Oct 2026)
- Bug fix after the freeze (24 Oct 2026, code only, no change to data, config or any definition): script 36 stopped with "non-finite function value" at target C = 0.70 because exp(beta * x) overflowed inside integrate() in censoring_rate_for() (R/secondary.R). Rewritten as plogis(beta * x - log mu), the same quantity; censoring rates for C = 0.60 and 0.65 are identical to before. Regression test added (test-secondary.R). Record in the deviations log as a code correction.
- Script 36 finished (24 Oct 2026). Power at alpha 0.05 for C = 0.60 / 0.65 / 0.70: GSE30219 0.53 / 0.87 / 1.00; GSE81089 0.54 / 0.92 / 1.00 (at the Holm step 0.05/8: 0.27 / 0.75 / 0.97); GSE72094 0.91 / 1.00 / 1.00; GSE50081 C = 0.70 1.00 (rest in power_simulation.csv).
- Speed fix after the freeze (24 Oct 2026, code only, numbers unchanged): pair_log_hr() (R/transport.R) copied the whole expression matrix (up to ~60k genes) for every pair in every bootstrap replicate, so script 33 ran > 30 min and script 34 would have taken hours. Now takes the signature genes before the resampled patients; scores bit-identical (checked), same random draws. Test added (test-null-and-transport.R).
- Second speed change (24 Oct 2026, code only): bootstrap replicates in

transport_bootstrap() now fit the Cox model with survival::coxph.fit using exactly the inputs coxph() passes (aeqSurv time fix, zero offset, Efron, nocenter = c(-1, 0, 1); checked against survival 3.8-13 source R/coxph.R lines 184, 486-494, 589-592), skipping formula handling; equality with coxph() to 1e-10 tested (with ties and near-ties). Progress messages (every 100 replicates, with time left) added to the bootstrap and per setting/pair to script 34, after script 33 was restarted several times because it printed nothing. Point SEs in script 33 still from coxph().

- Outcome analyses complete (24 Oct 2026; commit 1833625, pushed). Review in claude/v2-results-review.md. Added script 37_pooling_sensitivity.R (interval-method sensitivity; HK narrower than common effect for Yao, q = 0.19). Found 108 v1 files in results/ from the merge of the old GitHub repository (24 Oct 15:28) - to be moved to archive_v1/.
- Git check (24 Oct 2026): merge commit ef1a8ac ("Resolved merge conflicts by keeping local files") brought in the old v1 repository (229fc8f lineage). Since the freeze, the only changes to v2 files are the intended ones (R/secondary.R, R/transport.R, R/survival_metrics.R, R/meta.R, scripts/34, scripts/37, run_all.R, 3 test files); config/ and data/signatures/ untouched; no v1 file in R/, so scripts 33/34 used only v2 functions. Tracked v1 material: 108 results files, 47 scripts (scripts/R, backups, numbered v1 scripts), signatures/*.tsv, 4 data/processed files, HANDOVER/PENDING/STATUS.md, BenchMark3.Rproj, docs/gene_mapping.csv; stray copies: df.rds, results.rds, MANIFEST.txt, scripts/gdc_api.R, scripts/meta.R, scripts/test-gdc-api.R, scripts/scripts.Rproj, scripts/00_setup(1).R. cleanup_v1.sh moves all of it to archive_v1/ (tested on a mock copy: 190 moves, v2 scripts and outputs untouched); PDFs of notes to docs/notes/. docs/ (literature coding, reporting completeness, cohort_overlap.csv cited by independence.csv, v1 deviations log) kept in place.
- v1 cleanup done (24 Oct 2026, commit 8d942ad, pushed): 190 files moved to archive_v1/ (a case-only duplicate results/tables/table1_signatures.csv was dropped by git on the case-insensitive Mac file system; v1 file, history kept). Full test suite 260/260.
- Script 40_paper_outputs.R + R/paper.R (24 Oct 2026): manuscript tables and figures from v2 outputs only -> results/paper/ (tables: signatures, cohorts, independent pairs, aggregate, sensitivity transportability, sensitivity H1, 18 supplementary CSVs; figures: coverage heatmap, C-index TCGA vs independent, HR by signature, Yao 2021 forest (registered + modified HK), transportability across settings, H1 vs random-gene null, KM curves of the 26 independent pairs). Tested in webR on the real v2 result tables with mock cohorts for the cohort-level parts (median follow-up, KM). Added to run_all.R.
- Draft 5 written (24 Oct 2026; ManuscriptDraft5_v2.docx; notes in claude/draft5-notes.md). docs/deviations_v2.md added (decisions after Update 3: A1-A8 before the freeze, B1-B5 after; OSF Update 4 to file). Script 40 table_signatures now includes the expression scale stated in each paper.